Supplementary Material for Benchmarking Foundation Models on Exceptional Cases: Dataset Creation and Validation

A Dataset Collection and Preprocessing

A.1 Graphic Novels

We collected the graphic novels through web scraping and then segmented them panel by panel using automated Python scripts. We reviewed and excluded data entries that contained unevenly sized panels to maintain consistency in the dataset. This dataset allows us to evaluate the extent to which the FMs comprehends the story line. To ensure an accurate assessment, we eliminate all clues that provide information about the story line, including panel numbers and titles of the graphic novel as shown in Figure 12.

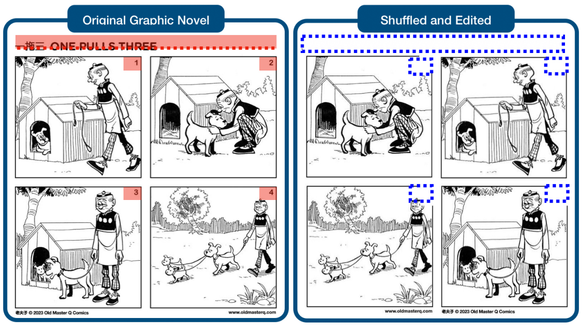


Figure 12: We remove clue-containing sections marked by red boxes that help determine the correct story line. These sections were removed as shown by the blue dotted line boxes in the 'Shuffled and Edited' version.

A.2 Calligraphy

We preprocessed the dataset according to three key rules. First, images were excluded if the resolution was too low or if the number of characters exceeded a threshold of 35, making them difficult for even humans to recognize. As illustrated in Figure 13, 35 characters represent an unusually large number in the dataset, and images with more than 35 characters were found to be visually challenging for human recognition. Therefore, we excluded images containing more than 35 characters from the evaluation. Second, images featuring multiple calligraphy pieces were separated using bounding boxes provided by the OCR API. Third, typographic elements, such as signs or watermarks irrelevant to the calligraphy, were cropped from the images. An example of the preprocessed Korean calligraphy dataset is provided in Figure 14.

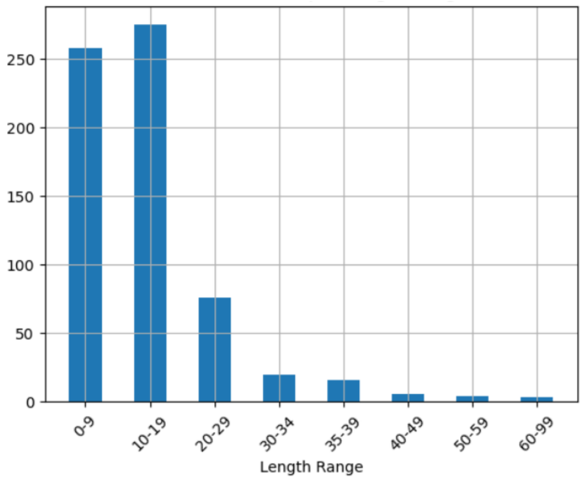


Figure 13: The bar plot of character lengths in Korean calligraphy images. We determined that images with over 35 characters presented considerable visual recognition difficulties, even for humans, prompting their exclusion from our evaluation.

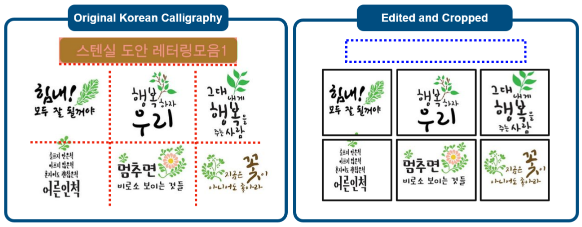


Figure 14: Example of preprocessed Korean calligraphy. We removed typographic elements unrelated to the calligraphy and automatically cropped overlapping sections using bounding boxes detected by the OCR API.

A.3 Onion, Not The Onion

We performed web scraping on The Onion website and Reddit's Not The Onion section. After data collection, we implemented a filtering process to enhance the quality of the dataset. Specifically, we removed instances where no content was collected, where duplicate content appeared, and where advertisements were included. During preprocessing, we encountered valid data of varying lengths, both long and short, which were indeed written by humans. These instances represent qualitative news articles, so we chose not to remove them to maintain the integrity of the dataset. As a result, the mean and median text lengths are 2,243 and 1,433 words, respectively, leading to a left-skewed distribution. A histogram illustrating text lengths for each data example, along with category-specific statistics, is presented in Figure 15 and Table 8. Through this process, we structured the dataset so that only the title and content of the original news articles influenced the FMs' judgment during the fake news detection task.

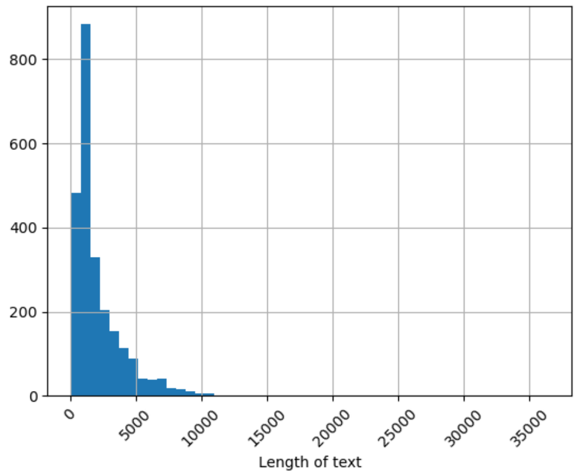


Figure 15: Length plot of the preprocessed Onion and Not the Onion news data.

	Overall	Onion	Not The Onion
Mean	2243	1048	3484
Med	1433	1115	2824
# Examples	2451	1249	1202

Table 8: Mean and Median length and # of examples of data. Compared to Not The Onion articles, the overall length of Onion articles is significantly shorter.

A.4 Lyrics

For the English dataset, we first collected the title and artist of each song, removing any duplicate entries where both the title and artist matched, as different songs can share the same title. Next, we generated links to the Genius website to retrieve song lyrics and descriptions. During this process, we removed strings that followed terms like "featuring" and adjusted characters, including brackets and Latin alphabets. In cases where song descriptions, genres, or lyrics were unavailable due to link generation errors or the lack of information on the site, the song was excluded from the dataset. Additionally, songs with non-English lyrics were removed. To avoid redundancy, any song that appeared in both the weekly and yearly datasets was excluded from the weekly dataset. For the genre detection task in English, we simplified the genre list by excluding genres with fewer than 10 occurrences, which resulted in a final set of 58 unique genres and a dataset of 1,811 songs. A similar process was applied to the Korean dataset, but with key differences: non-Korean lyrics were retained due to their high frequency, and no genre cleaning was performed, as the dataset contained fewer distinct genres. Importantly, no songs were excluded from the Korean dataset during the lyric, description, or genre retrieval process because all song data was sourced from Melon, unlike the English dataset, which pulled information from multiple websites. The number of remaining data points at each stage of the process is outlined in Table 9. For the infilling task, only the 2024 unseen dataset was utilized to address copyright safety concerns associated with Foundation Models (FMs). Entries that exceeded a BERT score threshold of 0.9 were excluded, as such high semantic similarity suggested the data lacked sufficient distinction or uniqueness.

	English		Korean	
	Seen	**Unseen**	**Seen**	**Unseen**
Total	3400	1700	3400	1700
Delete duplicate songs	3112	353	2187	304
Lyrics and Description crawling	2435	246	2187	304
Genre crawling	1828	139	2187	304
Remove Multilingual	1803	131	X	X
Remove duplicate between yearly and weekly	X	121	X	176
Cleaning Genre	1703	108	X	X
Final	1703	108	2187	176

Table 9: During the song data collection process, various criteria were applied to filter out certain songs, as outlined in the first column of the table. The numbers in each block represent the remaining data after each step, while an "X" indicates that the dataset did not undergo that particular step.

B Experiments Details

B.1 Graphic Novels

The API temperature setting is adjusted to 0.01 for Gemini-1.5-Pro and 0 for GPT-4o and Claude-3.5-Sonnet to ensure consistent results. We used SBERT model, with the 'all-mpnet-base-v2' pre-trained model configuration, to evaluate the accuracy of the FMs' descriptions in capturing the visual content of images. To generate a concise answer, the model is instructed to output the response solely in the format [1,2,3,4], as shown in the blue text in Figure 16 ('Prompt'). We set the ground truth order as [1,3,2,4] to automate the task, given

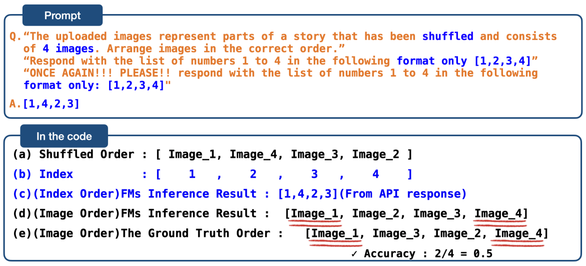


Figure 16: Description of the random shuffle experiment process: In the 'Prompt', all essential information is provided, including the fact that all images are shuffled, that the four images are parts of a story, and the response format. The 'Code' section illustrates the task sequence from (a) to (e). (a) shows the shuffled input image order, (b) is the index of the input image order, (c) is FMs' response which is the inferred result, (d) is the transformation from index order to image order, and (e) is the ground truth order used to calculate accuracy.

Graphic Novels	
Example	Prompt
ORCoT (Detailed Multi-Step Version)	Input : Q. "The uploaded images represent parts of a story that has been shuffled and consists of 4 images." "Arrange images in the correct order." IMPORTANT: Respond ONLY with the list of numbers 1 to 4 in this format: [1, 2, 3, 4]. A. Let's think step by step. 1. Initial Observation: Look at the comic image for a moment. What stands out to you immediately? 2. Setting Description: Describe the setting. Where does the scene take place? Include details about the background and environment. 3. Character Identification: Who are the characters in the image? Describe their appearance and any notable features. 4. Actions and Interactions: What are the characters doing? Describe their actions and how they interact with each other. 5. Text Elements: What text elements are present? What are the characters saying or thinking, and how does this contribute to the scene? 6. Emotional Tone and Atmosphere: What is the emotional tone of the scene? Describe the mood and emotions conveyed by the characters and setting. 7. Context and Story Progression: What do you think happened before this scene, and what might happen next? How does this image fit into the larger story? 8. Summary and Interpretation: Summarize your description. What is the key aspect of this comic image, and what theme or message does it convey? By these logical steps, the correct order of the images is: Output: A.

Table 10: We tried many other version of ORCoT to enhance capability of FMs on Graphic Novels dataset such as the prompt in this table.

that the input images are shuffled, as shown in (e) in Figure 16 ('In the code'). This predetermined order allows us to verify whether FMs produces the correct sequence. Additionally, we demonstrate how the prompts were designed for each style in E.1. Table 15. We design the random shuffle experiment as follow.

1. Inform the FMs that the uploaded images represent parts of a story that have been shuffled and consist of four images as shown in the blue letters in Figure 16 ('Prompt'). Instruct it to analyze all the images and deduce the correct sequence.
2. Upload four images in a shuffled order, with each image assigned an ID number as shown in (a), (b) in Figure 16 ('In the code').
3. The uploaded images are indexed, and the FMs infers the correct order, subsequently outputting the images in the proper indexed sequence as shown in (c) in Figure 16 ('In the code').
4. Using code, the indexed sequence is transformed into a sequence of image ID numbers to obtain the image order predicted by the FMs as shown in (d) in Figure 16 ('In the code').
5. Compare the predicted image order with the ground truth order to determine accuracy as shown in (e) in Figure 16 ('In the code').

B.2 Calligraphy

For further analysis, we investigate how sentence length influences the performance of FMs, as the calligraphy dataset includes sentences of varying lengths. To ensure a thorough evaluation, we assess multiple metrics, including precision, recall, and F1-score. Using these metrics, we perform experiments on our Korean calligraphy dataset. To achieve a detailed analysis, we examine the data across three distinct length categories: character-level, word-level, and the overall semantic meaning of the calligraphy.

B.3 Onion, Not The Onion

In this study, the default prompt instructed the model to directly distinguish between fake and real news. In contrast, the ORCoT prompts guided the model through a step-by-step reasoning process, specifically optimized for identifying exceptional cases of fake news. This approach required the

Character-Level		Zero-shot	ORCoT	ORCoT+Few-shot
Claude-3.5-Sonnet	Precision	22.20	23.65	29.54
	Recall	25.93	29.89	32.94
	F1-Score	23.92	26.41	31.15
Gemini-1.5-Pro	Precision	15.58	17.09	18.56
	Recall	19.12	21.15	22.81
	F1-Score	17.35	18.91	20.41
GPT-4o	Precision	75.46	76.27	77.25
	Recall	74.82	75.50	76.17
	F1-Score	74.51	75.29	76.14

Table 11: Result(%) of Korean Calligraphy OCR task in character-level

Word-Level		Zero-shot	ORCoT	ORCoT+Few-shot
Claude-3.5-Sonnet	Precision	17.54	28.81	40.90
	Recall	19.77	38.10	43.70
	F1-Score	17.53	32.81	42.26
Gemini-1.5-Pro	Precision	18.89	21.01	22.63
	Recall	21.81	24.67	25.40
	F1-Score	20.25	22.70	23.93
GPT-4o	Precision	84.57	86.47	87.48
	Recall	86.64	87.83	88.75
	F1-Score	85.59	87.14	88.11

Table 12: Result(%) of Korean Calligraphy OCR task in word-level

Overall Meaning	Zero-Shot	ORCoT	ORCoT+Few-Shot
Claude-3.5-Sonnet	0.686	0.747	0.811
Gemini-1.5-Pro	0.608	0.676	0.680
GPT-4o	0.838	0.857	0.861

Table 13: Cosine similarity scores of Korean Calligraphy OCR task

		Genre Classification	Description Generation	Lyrics Infilling
Korean	Before Cut-Off	- Overlap Ratio - Exact Match		
	After Cut-Off	- Overlap Ratio - Exact Match		- ROUGE - BERT Score
English	Before Cut-Off	- Overlap Ratio - Exact Match	- ROUGE - BERT Score	
	After Cut-Off	- Overlap Ratio - Exact Match	- ROUGE - BERT Score	- ROUGE - BERT Score

Table 14: Evaluation metrics for each *Lyrics* task are presented. An empty block indicates that the corresponding data was not used for that particular task.

model to follow a specific sequence of thought steps. To evaluate the model's performance, we used both Few-Shot and ORCoT prompts, providing examples of fake and real news along with explanations of the reasoning process. These comparisons allowed us to assess the impact of different prompting methods on the model's accuracy in recognizing fake news. Detailed prompts are provided in Table 17.

B.4 Lyrics

An exact match score assigns a value of '1' when the predicted genre aligns perfectly with the original genre. The overlap ratio, on the other hand, measures similarity by assessing the proportion of shared elements between the predicted and original genres.

C Additional Experimental Examples

C.1 Graphic Novels

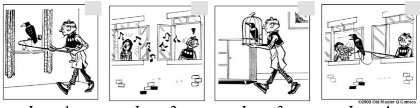
We conducted additional experiments on the graphic novel dataset, following the approach outlined in the Graphic Novels' *Qualitative Results section 4.1.2*. **In the first example** shown in Figure 17, All three models failed to determine the correct order. In particular, the descriptions generated by all FMs for *Image1* deviated significantly from the ground truth. For instance, GPT-4o described "something on a tray," which was not present in the image, Gemini-1.5-Pro mentioned "a crow perched on his shoulder," which did not align with the scene, and Claude-3.5-Sonnet identified "a perch," which was also absent from the image. **In the second example**, presented in Figure 18, all three FMs produced poor results. Both GPT-4o and Gemini-1.5-Pro generated an order with 0% accuracy, while Claude-3.5-Sonnet achieved 50% accuracy, indicating an overall failure to grasp the story conveyed in the images. For *Image2*, the descriptions generated by all FMs received low SBERT scores. In this image, a hunter encounters a tiger suffering from a thorn lodged in its paw; however, all models misrepresented the scene. For instance, GPT-4o mentioned a "calmer tiger," which did not align with the ground truth. **The third example** is a case that demonstrates properly inferring the correct order. As shown in Figure 19, considering that a low SBERT score was obtained for a single image, it can be inferred that even when Gemini-1.5-Pro and GPT-4o correctly determined the sequence, they did not achieve perfect recognition of all the images to determine the correct order. Claude-3.5-Sonnet incorrectly identified a peacock as a crane in *Image1*, resulting in a poor SBERT score. Additionally, FMs that received lower SBERT scores for *Image4* misrepresented the characters and provided insufficient details about the scene. For instance, Gemini-1.5-Pro incorrectly described a man as a woman and mentioned a third person who was not present in the image.

C.2 Calligraphy

We carried out additional experiments on the calligraphy dataset, following the methodology outlined in Calligraphy's *Qualitative Results section 4.2.2*. **The first example** of a calligraphy image depicts the word '춤' which means 'dance' in Korean. In this case, both Gemini-1.5-Pro and Claude-3.5-Sonnet produced poor results, while GPT-4o

Analysis of Example Input Results

[Shuffled Image Example Input]



[Random Shuffle Task Accuracy For The Example Input]

	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Shuffle Task Acc.(%)	25.00%	50.00%	25.00%

[SBERT Scores (Cosine-Similarity) For The Example Input]

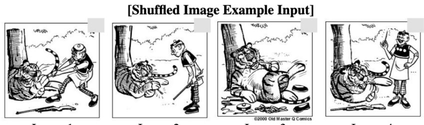
	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Image 1	0.307	0.528	0.495
Image 2	0.616	0.515	0.385
Image 3	0.466	0.576	0.765
Image 4	0.511	0.429	0.605

[Description For Image 1]
Ground Truth : The man has perched the crow on a stick and is using it to send the crow out of the window.
GPT-4o : The man offers the bird something on a tray.
Gemini-1.5-Pro : A man walks with a crow perched on his shoulder, holding a birdcage.
Claude-3.5-Sonnet : A person carries a perch with a crow sitting on it.

Figure 17: Overall SBERT Scores are mostly low for the example input, resulting poor accuracy result for shuffle task. Especially, all models illustrated the lowest SBERT score for *Image1*, where all models describe the image improperly.

Analysis of Example Input Results

[Shuffled Image Example Input]



[Random Shuffle Task Accuracy For The Example Input]

	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Shuffle Task Acc.(%)	0.00%	0.00%	50.00%

[SBERT Scores (Cosine-Similarity) For The Example Input]

	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Image 1	0.678	0.683	0.668
Image 2	0.463	0.479	0.495
Image 3	0.718	0.604	0.598
Image 4	0.384	0.638	0.648

[Description For Image 2]
Ground Truth : A hunter discovered a thorn lodged in the tiger's paw.
GPT-4o : The man observes the now calmer tiger.
Gemini-1.5-Pro : The hunter lures the tiger out with a ball.
Claude-3.5-Sonnet : The person approaches the resting tiger cautiously with a stick.

Figure 18: All three FMs performed poorly, suggesting a general inability to understand the story depicted in the images. For *Image2*, the descriptions generated by the FMs received low SBERT scores, as each model inaccurately represented the scene.

accurately recognized the word across all three prompt styles, as shown in Figure 20. Although Gemini-1.5-Pro correctly inferred the word in the ORCoT+Few-Shot prompt scenario, Claude-3.5-Sonnet failed to recognize it as a character across all three prompt styles. **The second example** of a calligraphy image illustrates the sentence '그는 나에게로 와 꽃이 되었다.' which translates to 'He came to me and became a flower.' in English. This sentence is part of a famous Korean poem. In this example, all models produced poor results, with GPT-4o's Zero-Shot response being the closest to matching the ground truth as shown in Figure 21. This is understandable, as the example comes from a poem, making its contextual meaning an exceptional case. FMs typically attempt to transcribe characters in calligraphy by relying on contextual understanding, as evidenced by the fact that, although incorrect, all other responses sounded quite natural. **The third example** of a calligraphy image illustrates the sentence '나로 오롯이 살아간다는 것' which translates to 'To live fully as myself.' in English. None of the models inferred the sentence correctly, possibly due to the character '나' (meaning 'myself') being printed unusually large to emphasize the core meaning of the sentence as shown in Figure 22. **The fourth example** of a calligraphy image features the word '산,' meaning 'Mountain' in English. As depicted in Figure 23, Gemini-1.5-Pro incorrectly identified it as a Chinese character, likely due to the style's resemblance to those popular in China. However, GPT-4o and Claude-3.5-Sonnet accurately described it in the 'ORCoT+Few-Shot' prompting, despite failing to infer it correctly in other prompting scenarios.

C.3 Onion, Not The Onion

We conducted an additional qualitative experiment to further explore the rationale behind the responses of a wider variety of FMs. In **the first example**, the article contains extraordinary claims, such as, "Adidas urgently recalled the German national team jersey featuring the number 44 due to its resemblance to symbols used by the German SS division". To verify these claims, the FMs underwent a validation process. Unlike the other FMs that correctly classified the article, GPT-4o failed to do so, despite using an internet search function for validation, as shown in Figure 24. **The second example** is a real news story from *Not The Onion*, reporting that a child in India escaped a monkey attack by using Amazon's Alexa. As shown in the Figure 25, Claude-3.5-Sonnet and Gemini-1.5-Pro correctly classified the story as real, while GPT-4o mistakenly classified it as fake news. Although GPT-4o followed appropriate reasoning steps, it generated the rationale that Alexa was not designed to produce barking sounds, leading it to deem the story illogical and subsequently classify it as fake. **The third example** is a fake news story from *Onion*, presenting a highly plausible narrative about Japan discovering 7,000 previously unknown

Analysis of Example Input Results

[Shuffled Image Example Input]

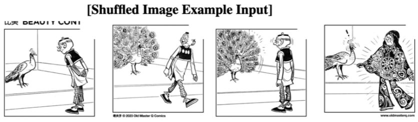


Image 1 Image 2 Image 3 Image 4

[Random Shuffle Task Accuracy For The Example Input]

	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Shuffle Task Acc.(%)	100.00%	100.00%	50.00%

[SBERT Scores (Cosine-Similarity) For The Example Input]

	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Image 1	0.875	0.682	0.531
Image 2	0.477	0.629	0.694
Image 3	0.882	0.828	0.815
Image 4	0.827	0.575	0.536

[Description For Image 4]

Ground Truth : The man, after seeing the peacock spread its feathers, seems to have remembered something and starts walking somewhere.

GPT-4o : he man returns, now dressed in a flashy, flamboyant outfit, outshining the peacock.

Gemini-1.5-Pro : A woman with elaborate patterns on her clothing outshines the peacock, capturing the man's attention.

Claude-3.5-Sonnet : The person, now wearing ornate clothing, stands proudly next to the bird. (Insufficient Information)

Figure 19: An example shows correct image sequence inference. However, the low SBERT score for a single image suggests that while Gemini-1.5-Pro and GPT-4o identified the overall sequence, they didn't perfectly recognize all images required for the correct order.

Comparison Between Prompts

Base Model	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Ground Truth	'춤'	'춤'	'춤'
Zero-Shot	'춤'	'ঽ'	No character
ORCot	'춤'	'꿈'	No character
ORCot + Few-Shot	'춤'	'춤'	No character

Figure 20: Gemini-1.5-Pro and Claude-3.5-Sonnet delivered suboptimal results, whereas GPT-4o accurately identified the word across all three prompt types. Although Gemini-1.5-Pro successfully inferred the word in the ORCoT+Few-Shot prompt scenario, Claude-3.5-Sonnet consistently failed to recognize it as a character in any of the three prompt styles.

islands. FMs produced varied responses to this story. As shown in the Figure 26, GPT-4o correctly identified the news as fake, reasoning that while the content was plausible and well-written, such a significant discovery would have been covered by major news outlets, but it was not. Claude-3.5-Sonnet initially classified it as real due to several credible-sounding pieces of evidence through Step 4: Cross-verification, but correctly reclassified it as fake in Step 5: Additional details, after noticing the inclusion of quotes from fictional individuals and unusual job titles. However, the scientific descriptions and seemingly credible sources in the article misled Gemini-1.5-Pro, causing it to classify the news as real.

Comparison Between Prompts

Base Model	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Ground Truth	'그는 나에게로 와 꽃이 되었다.'	'그는 나에게로 와 꽃이 되었다.'	'그는 나에게로 와 꽃이 되었다.'
Zero-Shot	'그는 나에게로 와서 꽃이 되었다'	'그는 네가 있어서 좋겠다'	'꽃 나눠 기쁨이 이루어지기를'
ORCot	'꽃길만 걸으세요 '	'오늘 당신에게 꽃이되겠습니다'	'꽃 나에게 오세요 이런사람'
ORCot + Few-Shot	'그는 나를 꽃으로 이름 지었다'	'꽃길만걸어요 '	'그는 나를 꽃다발로 이루었다'

Figure 21: In this case, all models performed poorly, except for GPT-4o's Zero-Shot response. This outcome is understandable, given that the example is drawn from a poem, making its contextual meaning particularly unique. Since FMs generally rely on contextual understanding to transcribe calligraphy characters, the other responses, though incorrect, still appeared fairly natural.

Comparison Between Prompts

Base Model	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Ground Truth	'나로 오롯이 살아간다는 것'	'나로 오롯이 살아간다는 것'	'나로 오롯이 살아간다는 것'
Zero-Shot	'로 모른이 살아간다는 것'	'무료이 갈아타는 날'	'로 오르이 살아간다는 것'
ORCot	'로 모든이 살아간다는 것'	'무릎이 시려서'	'론 오늘이 실아간다는 것'
ORCot + Few-Shot	'로 모든이 살아간다는 것'	'나를 찾는 시간'	'론 모르이 살아간다는 것'

Figure 22: The sentence was not correctly inferred by any of the models, possibly because the character '나' (meaning 'myself') was printed in an unusually large size to highlight its importance in the sentence.

C.4 Lyrics

We conducted an additional qualitative experiment to further explore a on infilling task, a song description generation task, and a genre classification task. In infilling task, as shown in **the first example** in the Figure 27, FMs generate a variety of incorrect responses. GPT-4 often misinterprets most of the masked segments, and it sometimes overlooks certain masked sections, leading to the omission of important tokens. Gemini-1.5-Pro exhibits problematic behaviors such as poor token inference, skipping token predictions entirely, and even deleting non-masked portions of the text (strikethrough in red). In **the second example**, as shown in Figure 28, from the English dataset, FMs exhibited better performance compared to the Korean dataset. Both FMs demonstrated a strong ability to predict tokens accurately for most of the masked parts. However, GPT-4o unexpectedly produced an inappropriate inference, predicting "my guard" instead of the more straightforward expression "my feelings.". In the song description generation task, the overall scores are poor, indicating that FMs struggles to understand the meaning of song lyrics. As demonstrated

Comparison Between Prompts

Base Model	GPT-4o	Gemini-1.5-Pro	Claude-3.5-Sonnet
Ground Truth	‘산’	‘산’	‘산’
Zero-Shot	‘중국’	‘書畫’	No character
ORCot	‘순’	‘春望’	No character
ORCot + Few-Shot	‘산’	‘春泥’	‘산’

Figure 23: Gemini-1.5-Pro mistakenly classified it as a Chinese character, likely because the style closely resembles those commonly seen in China. In contrast, GPT-4o and Claude-3.5-Sonnet provided accurate descriptions in the ’ORCoT+Few-Shot’ prompting, although they were unable to infer it correctly in other prompting scenarios.

Reasoning Step on Real News from Not The Onion (Article Length: Q1)

[GPT-4o]
Step 3. **Consistency and Accuracy**: The claim that Adidas would ban a number due to its resemblance to Nazi symbols is extraordinary and would likely be widely reported by major news outlets. The inconsistency in allowing certain names while banning others also seems improbable and lacks internal logic.
Inconsistent logic and lack of credible verification. (0) — GT (1)

[Claude-3.5-Sonnet]
Step 3. **Consistency and Accuracy**: The information presented seems consistent throughout the article. Given Germany's strict laws regarding Nazi symbols, this action by Adidas is plausible.
Plausible, needs verify (1) — GT (1)

[Gemini-1.5-Pro]
Step 3. **Consistency and Accuracy**: The article discusses a sensitive topic, and claims that Adidas has taken action to avoid any association with Nazi symbols. It mentions a specific decision regarding the number 44 on German national team jerseys and cites a well-known news source, Bild. These elements contribute to its credibility.
Rationale: Reported by reliable sources (1) — GT (1)

Figure 24: The first example of the FMs’ response to real news from the Not The Onion dataset illustrates part of the reasoning process in the fake news detection task. When dealing with extraordinary information, FMs make errors. For instance, GPT-4o mistakenly classified a true news story about Adidas and a Nazi symbol as fake, despite the ground truth being real.

Additional Experiment For Onion, Not The Onion

[GPT-4o]
Step 3. **Consistency and Accuracy**: Alexa is not designed for barking like a dog, which is technically implausible and unlikely to be a real feature. The story contains logical inconsistencies, such as why the monkey would immediately flee, making the article questionable. Sensationalized story, lacks verified sources or similar reports from multiple trusted outlets. (0) — GT (1)

[Claude-3.5-Sonnet]
Step 3. **Consistency and Accuracy**: The story describes a plausible scenario in India, where monkey intrusions are not uncommon. The use of Alexa to make dog barking sounds is a known feature. the story seems plausible and is reported by a credible source. Plausible but unverified (1) — GT (1)

[Gemini-1.5-Pro]
Step 3. **Consistency and Accuracy**: The events described - a monkey entering a house, a young girl using Alexa to scare it away - are entirely within the realm of possibility. Considering the reliable source, neutral language, consistent details, and the absence of any red flags, the story appears to be real. (1) — GT (1)

Figure 25: The second example is a real news story from *Not The Onion*. Claude-3.5-Sonnet and Gemini-1.5-Pro correctly identified the story as real, whereas GPT-4o incorrectly classified it as fake news. Despite following a logical reasoning process, GPT-4o concluded that Alexa was not programmed to produce barking sounds, which it found illogical, leading to the mistaken classification of the story as fake.

Additional Experiment For Onion, Not The Onion

[GPT-4o]
Step 4. **Cross-verification**: No reference is made to other news outlets or authoritative bodies that confirm this discovery. If this were true, it would have been widely covered by reliable sources such as government agencies or major news networks. (0) — GT (0)

[Claude-3.5-Sonnet]
Step 3. **Checking consistency and accuracy** - The claim about discovering 7,000 new islands due to improved surveying technology is plausible.
Step 4. **Cross-verification**: This news was actually reported by multiple reputable sources in early 2023, including CNN and The Guardian.
Step 5. **Additional details**: The article includes quotes from fictional people with unusual job titles, which is characteristic of satirical news. Rationale: Satirical twist on real news (0) — GT (0)

[Gemini-1.5-Pro]
Step 3. **Checking consistency and accuracy** - The claim of new islands discovered due to advanced technology is plausible and aligns with scientific advancements.
Step 4. **Cross-verification**: Further research would be needed to confirm if other reliable sources have reported on this discovery. Final output: Reliable source (1) — GT (0)

Figure 26: The third example is a fake news story from The *Onion*, and the FMs produced differing responses. GPT-4o correctly identified the news as fake, while Claude-3.5-Sonnet initially classified it as real, influenced by several credible-sounding details. However, in Step 5: Additional details, it correctly reclassified the story as fake. Gemini-1.5-Pro, misled by the scientific descriptions and seemingly credible sources in the article, incorrectly categorized the news as real.

in the Figure 29, the song discusses ’enduring difficult times with loved ones,’ whereas FMs describes it as ’dealing with a problematic relationship and addictive emotions. The similarity in the responses from all three FMs suggests that they simply summarized and repeated the input instead of offering a more in-depth interpretation. Also, This might be because non-lyrical elements, such as information about singer, were included in the ground truth due to the lack of specific format. Therefore we used F1 score as the primary evaluation metric that can assess if the generated description included or was included in the original description. In the genre classification task, as shown in **the first example** in the Figure 30, none of the FMs accurately matched the ground truth. The genre was broadly categorized as "rock," with sub-genres of "pop" and "folk." However, across all prompt styles, the FMs classified "pop" as the main genre. In **the second example**, while the FMs correctly identified "pop" as the main genre across all prompt scenarios(in orange), none accurately recognized the sub-genre "electro pop," which is the ground truth as shown in Figure 31.

(Korean) Comparison of Infilling Task Outputs Across FMs

Masked Part

[Ground Truth]

네가 참 궁금해 그건 너도 마찬가지 이거면 충분해 쫓고 쫓는 이런 놀이 참을 수 없는 이끌림과 호기심 묘한 너와 나 두고 보면 알겠지 Woo 눈동자 아래로 Woo 감추고 있는 거 Woo yeah It's so bad It's good 난 그 맘을 좀 봐야겠어 Narcissistic, my god I love it 서로를 비춘 밤 아름다운 까만 눈빛 더 빠져 깊이

[GPT-4o Response]

난 참 궁금해 미치겠어 (Blank) (Blank) (Blank) 이거면 충분해 (Blank) 쫓는 이런 느낌 참을 수 없는 이끌림과 호기심 묘한 너와 나 둘이 보면 알겠지 Woo 눈동자 아래로 Woo 감추고 있는 거 Woo yeah 너무 so bad 우린 good 난 그 맘을 좀 봐야겠어 Narcissistic, my god I love it 서로를 탐하는 밤 아름다운 까만 눈빛 더 빠져 깊이

[Gemini-1.5-Pro Response]

넌 참 궁금해 어쩌면 완벽해 이대로 완벽해 ~~이거면 충분해~~ (널) 쫓는 이런 마음 참을 수 없는 이끌림과 호기심 묘한 너와 나 만나 보면 알겠지 Woo 눈동자 아래로 Woo 감추고 싶어도 Woo yeah It's so bad It's (so) good 너의 그 맘을 좀 봐야겠어 Narcissistic, my god I love it 숨 (참고) love (dive) 밤 아름다운 까만 눈빛 더 빠져 깊이

Figure 27: FMs produce a range of incorrect responses. GPT-4 frequently misinterprets the masked segments, occasionally neglecting some masked sections, which results in the omission of crucial tokens. Gemini-1.5-Pro displays issues such as inaccurate token inference, skipping token predictions altogether, and even removing unmasked portions of the text (strikethrough in red).

(English) Comparison of Infilling Task Outputs Across FMs

Masked Part

[Ground Truth]

It's a light relief from a bad habit It's my mother's cookin' when I can't have it It's the last train home from the day trippin' It's the place I know when I start slippin' Darling, won't you take me home? Send me shivers somewhere I used to go Wrap my name across your mouth When I let my feelings down Darling, won't you take me home?

[GPT-4o Response]

It's a light relief from a bad habit Like my mama's cookin' when I can't have it It's the last train home from a day trippin' It's the place you know when you start slippin' Darling, won't you take me home? Send me shivers somewhere I used to go Write my name across your mouth When I let my guard down Darling, won't you take me home?

[Gemini-1.5-Pro Response]

It's a light relief from a bad habit Like my mama cookin' when I can't hack it It's the last train home from a day trippin' It's the place I know when I start slippin' Darling, won't you take me home? Send me shivers somewhere I used to go Wrap my name across your mouth When I let my feelings down Darling, won't you take me home?

Figure 28: In the English dataset, FMs showed improved performance relative to the Korean dataset. Both models were generally effective in accurately predicting tokens for most masked sections. However, GPT-4o made an unexpected error, generating the phrase "my guard" rather than the simpler and more appropriate expression "my feelings.".

Comparing Song Description Generation Output

[Ground Truth]

"Lose Control" is the second track on I've Tried Everything But Therapy by Teddy Swims. Swims uses this song to take everything off his shoulders- losing himself and all control when everything starts caving in; the only way to escape is being with that specific person.

➡️ **Song about Enduring difficult times with loved ones**

[GPT-4o Output]

"Lose Control" by Nessa Barrett, The lyrics convey a sense of losing oneself and falling apart when the partner is not around, highlighting the addictive and problematic nature of their connection.

[Claude-3.5-Sonnet Output]

"Lose Control" by an unspecified artist, The singer confesses to feeling vulnerable when separated from their lover. They describe their desire as addictive and problematic.

[Gemini-1.5-Pro Output]

"Lose Control" by MEDUZA and Becky Hill featuring Goodboys is about the singer's turbulent relationship with someone who makes them feel unstable and out of control, yet they crave the intense connection they feel when they're together.

➡️ **Song about Dealing with a problematic relationship and addictive emotions**

Figure 29: In the description generation task, the FMs misunderstood the song lyrics. Unlike the ground truth, all FMs described "dealing with a problematic relationship and addictive emotions." Their similar responses suggest they summarized and repeated the input rather than providing deeper interpretation.

Comparison of Genre Classification Task Outputs Across FMs

Title	GT	Base Model	Zero-Shot	ORCoT	ORCoT+Few-Shot
Beautiful Things	['folk rock', 'pop rock']	Claude-3.5-Sonnet	[country, country pop, americana]	[country, country pop, americana, folk]	[pop, country pop, folk pop]
		Gemini-1.5-Pro	[Country Pop]	[pop, r&b]	[country, country pop]
		GPT-4o	[pop, country pop, folk pop]	[country, country pop]	[country, country pop]

Figure 30: None of the FMs accurately matched the ground truth. The genre was broadly categorized as "rock," with "pop" and "folk" as sub-genres. However, in all prompt styles, the FMs misclassified "pop" as the main genre.

Comparison of Genre Classification Task Outputs Across FMs

Title	GT	Base Model	Zero-Shot	ORCoT	ORCoT+Few-Shot
Is It Over Now	['electro pop']	Claude-3.5-Sonnet	[pop, alternative pop, pop rock]	[pop, alternative pop, pop rock]	[pop, alternative pop, indie pop]
		Gemini-1.5-Pro	[pop]	[pop]	[pop, r&b]
		GPT-4o	[pop, alternative pop]	[alternative pop, pop rock]	[pop, alternative pop]

Figure 31: While the FMs correctly identified "pop" as the main genre across all prompt scenarios (highlighted in orange), none of the models accurately recognized the sub-genre "electro pop," which is the ground truth.

D Prompts

D.1 Graphic Novels

Graphic Novels	
Example	**Prompt**
Zero-Shot	Input : “The uploaded images represent parts of a story that has been shuffled and consists of 4 images." "Arrange images in the correct order.” “Respond with the list of numbers 1 to 4 in the following format only [1,2,3,4]” “ONCE AGAIN!!! PLEASE!! respond with the list of numbers 1 to 4 in the following format only: [1,2,3,4]" (Task Images) Output: A.
ORCoT + Zero-Shot	Input : Q. “The uploaded images represent parts of a story that has been shuffled and consists of 4 images." "Arrange images in the correct order.” IMPORTANT: Respond ONLY with the list of numbers 1 to 4 in this format: [1, 2, 3, 4]. (Task Images) Output: A. Let’s think step by step. The correct order is
ORCoT + Few-Shot	Input : Q. “The uploaded images represent parts of a story that has been shuffled and consists of 4 images." "Arrange images in the correct order.” IMPORTANT: Respond ONLY with the list of numbers 1 to 4 in this format: [1, 2, 3, 4]. **“The First, Example:”**: (1st Example Images) A. “Let’s think step by step. The correct order is [1,2,3,4]” **“The Second, Example:”**: (2nd Example Images) A. “Let’s think step by step. The correct order is [1,2,3,4]” **“The Third, Example:”**: (3rd Example Images) A. “Let’s think step by step. The correct order is [1,2,3,4]” Q. “The uploaded images represent parts of a story that has been shuffled and consists of 4 images." "Arrange images in the correct order.” IMPORTANT: Respond ONLY with the list of numbers 1 to 4 in this format: [1, 2, 3, 4]. (Task Images) Output: A. Let’s think step by step. The correct order is

Table 15: We assigned a response format to the Foundation Models (FMs) twice because, in the Zero-Shot setting, the variation in responses was too broad, causing the FMs to occasionally deviate from the response format. In the ORCoT+Few-Shot setting, we used three different examples, as performance was insufficient with only one or two examples. In this table, we refer to the simpler CoT format, as ORCoT’s prompt length is too long. For the full ORCoT prompts, please refer to Table 10.

D.2 Calligraphy

Dataset Name	
Example	**Prompt**
Zero-Shot	Input : One korean calligraphy image Prompt : "What are the all Korean characters in the image? Make sure that your answer only includes the result of the OCR without translating. You don’t need to describe the processing steps." Output: Only OCR result text
ORCoT + Zero-Shot	Input : One korean calligraphy image Prompt : "The image uploaded is Korean calligraphy with illustration. Transcribe the letters in the uploaded image. Solve it with following steps. 1. Identify the start and end of the sentence. Check if there are any line breaks in the middle of the sentence. 2. Split the recognized text into individual words. Combine the split words based on the context to form a coherent sentence. 3. Analyze the context to infer the meaning of the handwriting. Correct typos by comparing them with similar words and choosing the correct one. 4. Perform grammar and spelling checks to verify the recognized sentence. Ensure that the sentence flows naturally and makes sense. Don’t describe your steps. Just answer the result of the OCR without translating." Output: Only OCR Result text
ORCoT + Few-Shot	Input : One korean calligraphy image Prompt : "Below are examples of OCR task. I’ll show image first and explain step-by-step how to extract text from the image." **Example1**: example1 image "Step1: Identify the start and end of the sentence. Check if there are any line breaks in the middle of the sentence. Identify that the sentence starts with ’바라는게’ and ends with ’안그래?’ Step2: Split into words and translate each word in English and identify any typos based on the context.: 바라는게 (What I hope for) 무한정 (infinitely) 끝없이 (endlessly) 내리는 (falling) 게 (particle, indicating ’is’) 아닌게 (is not) 엄마나 (Typo: misidentified word, Correct: 얼마나, Translation: how much) 다행인지 (fortunately) 몰라 (I don’t know) 안그래? (isn’t it?) Step3: Correct the typos by comparing each word with similar words and combine the corrected words to form a coherent sentence.: ’엄마나’ should be ’얼마나’, ’알고 래?’ should be ’안그래?’ Step4: Combine based on context: ’바라는게 무한정 끝없이 내리는 게 아닌게 얼마나 다행인지 몰라 안그래?’ There is no weird word to use. Step5: Analyze the context to infer the meaning of the handwriting. Correct any misrecognized words by comparing them with similar words and choosing the correct one. Infer the context: The sentence talks about how fortunate it is that something is not happening endlessly. Correct any misrecognized words: ’얼 마나’ should be ’얼마나’ Step6: Perform grammar and spelling checks to verify the recognized sentence. Ensure that the sentence flows naturally and makes sense. Check grammar and spelling: Ensure ’바라는게 무한정 끝없이 내리는 게 아닌게 얼마나 다행인지 몰라 안그래?’ is grammatically correct and makes sense. Ensure the sentence flows naturally and the meaning is clear." prompt: "Now, please perform an OCR task on the following image like the example. The image is Korean calligraphy with an illustration. Transcribe the letters in the picture with a step-by-step explanation of your reasoning. But Don’t describe your steps. Just answer the result of the OCR without translating." Output: Only OCR Result text

Table 16: Korean Calligraphy Prompt: For the ORCot+Few shot prompt, We utilized two examples but only one example is listed in the paper because it was too long to attach. The full prompt can be seen in GitHub.

D.3 Onion Not The Onion

Onion, Not The Onion	
Example	Prompt
Zero Shot	Input : A News article and Title Prompt: The uploaded text is one of the articles that may be real or fake. Please Answer whether below article is fake or real. Say nothing but **the number 0 or 1**. i.e. Answer 1 if you think the article is real, answer 0 if you think it is fake Output: (0 ‖ 1)
ORCoT + Zero Shot	Input : A News article and Title The uploaded text is one of the articles that may be real or fake.Please Answer whether below article is fake or real. Give a 20-character rationale for why you think that way, and output a 0 and 1 at the end of the sentence. To Solve this, You have to think step by step. **The first step** in identifying fake news is evaluating the reliability of the information source. Well-known and verified news organizations are generally more reliable, and their reports can be trusted more than unverified sources. **In addition** to source reliability, look at the language used in the content. Fake news often uses sensational or exaggerated language designed to elicit an emotional response. **It is also important to check** for consistency and accuracy in the information presented; fake news typically includes claims that are either unverified or clearly false. **Another critical step** is cross-verification,where check if the same claims are reported by multiple trusted sources. i.e. rationale + answer 1 if you think the article is real, rationale + answer 0 if you think it is fake. Must Keep in mind that the end of a sentence should end with either 0 or 1 Output: (rationales + (0‖1))
ORCoT + few Shot	Input : A News article and Title The uploaded text is one of the articles that may be real or fake.Please Answer whether below article is fake or real. Give a 20-character rationale for why you think that way, and output a 0 and 1 at the end of the sentence. To Solve this, You have to think step by step. **The first step** in identifying fake news is evaluating the reliability of the information source. Well-known and verified news organizations are generally more reliable, and their reports can be trusted more than unverified sources. **In addition** to source reliability, look at the language used in the content. Fake news often uses sensational or exaggerated language designed to elicit an emotional response. **It is also important to check** for consistency and accuracy in the information presented; fake news typically includes claims that are either unverified or clearly false. **Another critical step** is cross-verification,where check if the same claims are reported by multiple trusted sources. **See the example below**. i.e. rationale + answer 1 if you think the article is real, rationale + answer 0 if you think it is fake. Must Keep in mind that the end of a sentence should end with either 0 or 1 **Example**: we provided one fake news story from The Onion and one real news story from Reddit's Not the Onion. Additionally, rather than merely presenting the news, **we included examples of the rationales we derived for the two news stories, following the same prompting method**. Output: (rationales + (0‖1))

Table 17: We provided examples of prompts used to detect fake news, focusing on the implementation of ORCoT reasoning. We presented a structured approach that outlines the steps a FMs considers when analyzing and concluding whether a news story is fake or real. Lastly, this method involves a few-shot learning technique where examples of fake news and real news are given alongside rationales.

D.4 Lyrics

D.4.1 English Genre Classification

Lyrics	
Example	Prompt
Zero-Shot	Input : Lyrics Prompt : Here is a list of unique music genres: ['genre list str']. Say nothing but the Genre as Genre: the output. Output example: Genre: [pop, r&b, hip hop]. Lyrics: 'lyrics' Output: Genre: the output
ORCoT + Zero-Shot	Input : Lyrics Prompt : Here is a list of unique music genres: ['genre list str']. Based on the lyrics provided, identify the genres. Say nothing but the Genre as Genre: the output. Output example: Genre: [pop, r&b, hip hop]. Lyrics: 'lyrics' Output: Genre: the output
ORCoT + Few-Shot	Input : Lyrics Prompt : Here is a list of unique music genres: ['genre list str']. **Example Lyrics**: And she spoke words that would melt in your hands And she spoke words of wisdom To the basement, people, to the basement Many surprises await you In the basement, people, in the basement You hid there last time, you know we're gonna find you Sick in the car seat, 'cause you're not up to going Out on the main streets, completing your mission You hid there last time, you know we're gonna find you Sick in the car seat, 'cause you're not up to going Out on the main streets, completing your mission **Example Description**: indie pop Now, based on the lyrics provided, identify the genres. Say nothing but the Genre as Genre: the output. Output example: Genre: [pop, r&b, hip hop]. Lyrics: 'lyrics' Output: Genre: the output

Table 18: Prompt for English genre classification task

D.4.2 Korean Genre Classification

Lyrics	
Example	Prompt
Zero-Shot	Input : Lyrics Prompt : Here is a list of unique music genres: [’genre list str’]. Say nothing but the Genre as Genre: the output. Output example: Genre: [발라드, 댄스, 랩/힙합]. Lyrics: ’lyrics’ Output: Genre: the output
ORCoT + Zero-Shot	Input : Lyrics Prompt : Here is a list of unique music genres: [’genre list str’]. Based on the lyrics provided, identify the genres. Say nothing but the Genre as Genre: the output. Output example: Genre: [발라드, 댄스, 랩/힙합]. Lyrics: ’lyrics’ Output: Genre: the output
ORCoT + Few-Shot	Input : Lyrics Prompt : Here is a list of unique music genres: [’genre list str’]. **Example Lyrics**: 처음 그대 내게로 오던 그날에 잠시 동안 적시는 그런 비가 아니길 간절히 난 바래왔었죠 그대도 내 맘 아나요 매일 그대만 그려왔던 나를 오늘도 내 맘에 스며들죠 그대는 선물입니다 하늘이 내려준 홀로 선 세상 속에 그댈 지켜줄게요 어느 날 문득 소나기처럼 내린 그대지만 오늘도 불러 봅니다 내겐 소중한 사람 Oh 떨어지는 빗물이 어느새 날 깨우고 그대 생각에 잠겨요 이제는 내게로 와요 언제나처럼 기다리고 있죠 그대 손을 꼭 잡아줄게요’ **Example Description**: 발라드, 국내드라마 Now, based on the lyrics provided, identify the genres. Say nothing but the Genre as Genre: the output. Output example: Genre: [발라드, 댄스, 랩/힙합]. Lyrics: ’lyrics’ Output: Genre: the output

Table 19: Prompt for Korean genre classification task

D.4.3 English Song Description Generation

Lyrics	
Example	Prompt
Zero-Shot	Input : Lyrics Prompt : Say nothing but the Description as Description: the output Output example: Description: The song explores themes of love and heartbreak. Lyrics: 'lyrics' Output: Description: the output
ORCoT + Zero-Shot	Input : Lyrics Prompt : Based on the provided lyrics, write a brief description of the song. Include the possible song title and artist name in the description. Say nothing but the Description as Description: the output Output example: Description: Honeymoon Avenue by Ariana Grande is about knowing you are at the end of a relationship and wishing it could not be the end and go back to the beginning and start over. Output: Description: the output
ORCoT + Few-Shot	Input : Lyrics Prompt : **Example Lyrics**: I'd like to say we gave it a try I'd like to blame it all on life Maybe we just weren't right But that's a lie, that's a lie And we can deny it as much as we want But in time, our feelings will show 'Cause sooner or later, we'll wonder why we gave up The truth is everyone knows, oh Almost, almost is never enough So close to being in love If I would have known that you wanted me the way I wanted you Then maybe we wouldn't be two worlds apart (Ah) But right here in each other's arms And we almost, we almost knew what love was But almost is never enough (Ah) If I could change the world overnight (Ah) There'd be no such thing as goodbye (Ah) You'd be standing right where you were (Ah) And we'd get the chance we deserve, oh (Ah) See upcoming pop shows Get tickets for your favorite artists Try to deny it as much as you want But in time, our feelings will show (Ah) 'Cause sooner or later, we'll wonder why we gave up The truth is everyone knows (Ah) **Example Description**: On the collaborative track "Almost Is Never Enough," Ariana Grande & Nathan Sykes play a couple who had a relationship that hadn't gone right. Ariana would like to say things were going well but she knows that's a lie and like the title states, almost is never enough to make the relationship work; you need to put full effort in. Both of them state that they didn't feel the relationship while in it, but the mood of the song and lyrics suggest that they both want to either reconnect or they simply just miss better times. At the time of the song's release, Nathan and Ariana were dating. Unfortunately, their relationship ended a few months later. Now, based on the provided lyrics, write a brief description of the song. Include the possible song title and artist name in the description. Say nothing but the Description as Description: the output Output example: Description: Honeymoon Avenue by Ariana Grande is about knowing you are at the end of a relationship and wishing it could not be the end and go back to the beginning and start over. Output: Description: the output

Table 20: Prompt for English song description generation task

D.4.4 English Song Infilling

Lyrics Infilling Task	
Example	Prompt
Zero-Shot	Input : Masked lyrics Prompt : You are a powerful language model. Fill in the blanks in the following text with appropriate words. The text is a part of a song with certain words masked by [MASK]. Lyrics: 'lyrics Say nothing but the filled lyrics as 'Filled lyrics: the output'. Output example: Filled lyrics: 'I know this pain (I know this pain) why do you lock yourself up in these chains? (these chains)... Output: Filled lyrics: the output
ORCoT + Zero-Shot	Input : Lyrics Prompt : You are a powerful language model. Fill in the blanks in the following text with appropriate words.The text is a part of a song with certain words masked by [MASK]. For each blank, think step by step about the context and meaning of the surrounding text before choosing the word. To do this, follow these steps: a. Carefully read and analysis the lyrics. b-1. Check the entire lyrics to see if there are any repeating parts. b-2. If repeating parts exist, replace the [MASK] with the corresponding word. c-1. Make the list of possible words for the masked part. c-2. Select a suitable word from the candidate list. c-3. Replace [MASK] with the word that you selected. Lyrics: 'lyrics Step-by-step reasoning and filled lyrics as 'Filled lyrics: the output'. Say nothing but the filled lyrics as 'Filled lyrics: the output'. Output example: Filled lyrics: 'I know this pain (I know this pain) why do you lock yourself up in these chains? (these chains)... Output: Filled lyrics: the output
ORCoT + Few-Shot	Input : Lyrics Prompt : You are a powerful language model. Fill in the blanks in the following text with appropriate words. The text is a part of a song with certain words masked by [MASK]. For each blank, think step by step about the context and meaning of the surrounding text before choosing the word. To do this, follow these steps: a. Carefully read and analysis the lyrics. b-1. Check the entire lyrics to see if there are any repeating parts. b-2. If repeating parts exist, replace the [MASK] with the corresponding word. c-1. Make the list of possible words for the masked part. c-2. Select a suitable word from the candidate list. c-3. Replace [MASK] with the word that you selected. Example: Lyrics: Rotgut whiskey's gonna ease my mind Beach [MASK] rests on the dryin' line Do I remind you of your daddy in his '88 Ford? Labrador [MASK] out the passenger door The sand from your hair is blowin' in my eyes [MASK] it on [MASK] [MASK] grown men don't cry [MASK] [MASK] remember that beat down basement couch? I'd sing [MASK] my love songs [MASK] you'd tell me about How your mama [MASK] off and pawned her ring [MASK] remember, I remember everything Filled lyrics: Rotgut whiskey's gonna ease my mind Beach towel rests on the dryin' line Do I remind you of your daddy in his '88 Ford? Labrador hangin' out the passenger door The sand from your hair is blowin' in my eyes Blame it on the beach, grown men don't cry Do you remember that beat down basement couch? I'd sing you my love songs and you'd tell me about How your mama ran off and pawned her ring I remember, I remember everything Now, based on the provided lyrics, fill in the blanks with appropriate words. Lyrics: 'lyrics Step-by-step reasoning and filled lyrics as 'Filled lyrics: the output'. Say nothing but the filled lyrics as 'Filled lyrics: the output'. Output example: Filled lyrics: 'I know this pain (I know this pain) why do you lock yourself up in these chains? (these chains)... Output: Filled lyrics: the output

Table 21: Prompt for English lyrics infilling task. Examples in ORCoT+Few-shot are composed of data removed during BERT testing.

D.4.5 Korean Song Infilling task

Lyrics	
Example	Prompt
Zero-Shot	Input : Masked lyrics Prompt : You are a powerful language model. Fill in the blanks in the following text with appropriate words. The text is a part of a song with certain words masked by [MASK]. Lyrics: 'lyrics Say nothing but the filled lyrics as 'Filled lyrics: the output'. Output example: Filled lyrics: 'I know this pain (I know this pain) why do you lock yourself up in these chains? (these chains)... Output: Filled lyrics: the output
ORCoT + Zero-Shot	Input : Lyrics Prompt : You are a powerful language model. Fill in the blanks in the following text with appropriate words. The text is a part of a song with certain words masked by [MASK]. For each blank, think step by step about the context and meaning of the surrounding text before choosing the word. To do this, follow these steps: a. Carefully read and analysis the lyrics. b-1. Check the entire lyrics to see if there are any repeating parts. b-2. If repeating parts exist, replace the [MASK] with the corresponding word. c-1. Make the list of possible words for the masked part. c-2. Select a suitable word from the candidate list. c-3. Replace [MASK] with the word that you selected. Lyrics: 'lyrics Step-by-step reasoning and filled lyrics as 'Filled lyrics: the output'. Say nothing but the filled lyrics as 'Filled lyrics: the output'. Output example: Filled lyrics: 'I know this pain (I know this pain) why do you lock yourself up in these chains? (these chains)... Output: Filled lyrics: the output
ORCoT + Few-Shot	Input : Lyrics Prompt : You are a powerful language model. Fill in the blanks in the following text with appropriate words. The text is a part of a song with certain words masked by [MASK]. For each blank, think step by step about the context and meaning of the surrounding text before choosing the word. To do this, follow these steps: a. Carefully read and analysis the lyrics. b-1. Check the entire lyrics to see if there are any repeating parts. b-2. If repeating parts exist, replace the [MASK] with the corresponding word. c-1. Make the list of possible words for the masked part. c-2. Select a suitable word from the candidate list. c-3. Replace [MASK] with the word that you selected. Example: Lyrics: 세상에 음악의 신이 있다면 고맙다고 안아주고 싶어 전 세계 공통의 Language 자음과 모음이 달라도 상관없는 건 Music 말이 안 통해도 [MASK] 있다면 [MASK] 지금부터는 아주 친한 친구 너와 내가 모르는 사이여도 춤출 [MASK] 있어 We [MASK] mix it up right Sugar and spice Brass sound and guitar 네 [MASK] 다 내 [MASK] 쿵치팍치 또한 내 이름인가 이것 또한 나를 위한 소린가 [MASK] [MASK] Drum bass Piano [MASK] Filled lyrics: 세상에 음악의 신이 있다면 고맙다고 안아주고 싶어 전 세계 공통의 Language 자음과 모음이 달라도 상관없는 건 Music 말이 안 통해도 음악이 있다면 우리는 지금부터는 아주 친한 친구 너와 내가 모르는 사이여도 춤출 수 있어 We can mix it up right Sugar and spice Brass sound and guitar 네 글자면 다 내 이름이래 쿵치팍치 또한 내 이름인가 이것 또한 나를 위한 소린가 Kick snare Drum bass Piano Bassline Lyrics: 'lyrics Step-by-step reasoning and filled lyrics as 'Filled lyrics: the output'. Say nothing but the filled lyrics as 'Filled lyrics: the output'. Output example: Filled lyrics: 'I know this pain (I know this pain) why do you lock yourself up in these chains? (these chains)... Output: Filled lyrics: the output

Table 22: Prompt for Korean lyrics infilling task. Examples in ORCoT+Few-shot are composed of data removed during BERT testing.